

Global Youtube Statistics-2022

```
In [33]: # importing the libraries;
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading dataset;

```
In [34]: # import dataset;
df = pd.read_csv(r'C:\Users\WINDOWS 10\Documents\Datasets\global\Global YouT
print(df)
```

	Rank	Youtuber	Subscribers	Video Views	\
0	1	T-Series	245000000	2.280000e+11	
1	2	YouTube Movies	170000000	0.000000e+00	
2	3	MrBeast	166000000	2.836884e+10	
3	4	Cocomelon - Nursery Rhymes	162000000	1.640000e+11	
4	5	SET India	159000000	1.480000e+11	
..	...	...	...	...	
990	991	Natan por Aïç	12300000	9.029610e+09	
991	992	Free Fire India Official	12300000	1.674410e+09	
992	993	Panda	12300000	2.214684e+09	
993	994	RobTopGames	12300000	3.741235e+08	
994	995	Make Joke Of	12300000	2.129774e+09	

	Category	Title	Uploads	Countr
y \				
0	Music	T-Series	20082	Indi
a				
1	Film & Animation	youtubemovies	1	United State
s				
2	Education	MrBeast	744	United State

Understanding the dataset;

```
In [35]: # print top 5 rows;
df.head()
```

Out[35]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Count
0	1	T-Series	245000000	2.280000e+11	Music	T-Series	20082	Ind
1	2	YouTube Movies	170000000	0.000000e+00	Film & Animation	youtubemovies	1	United State
2	3	MrBeast	166000000	2.836884e+10	Entertainment	MrBeast	741	United State
3	4	Cocomelon - Nursery Rhymes	162000000	1.640000e+11	Education	Cocomelon - Nursery Rhymes	966	United State
4	5	SET India	159000000	1.480000e+11	Shows	SET India	116536	Ind

5 rows × 28 columns



In [36]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 28 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Rank                                                                 995 non-null   int64
1   Youtuber                                                            995 non-null   object
2   Subscribers                                                         995 non-null   int64
3   Video Views                                                         995 non-null   float64
4   Category                                                            949 non-null   object
5   Title                                                               995 non-null   object
6   Uploads                                                             995 non-null   int64
7   Country                                                             873 non-null   object
8   Abbreviation                                                        873 non-null   object
9   Channel Type                                                        965 non-null   object
10  Video Views Rank                                                    994 non-null   float64
11  Country Rank                                                         879 non-null   float64
12  Channel Type Rank                                                    962 non-null   float64
13  Video Views for the last 30 days                                     939 non-null   float64
14  Lowest Monthly Earnings                                             995 non-null   float64
15  Highest Monthly Earnings                                            995 non-null   float64
16  Lowest Yearly Earnings                                              995 non-null   float64
17  Highest Yearly Earnings                                             995 non-null   float64
18  Subscribers for last 30 days                                         658 non-null   float64
19  Created Year                                                         990 non-null   float64
20  Created Month                                                        990 non-null   object
21  Created Date                                                         990 non-null   float64
22  Gross Tertiary Education Enrollment (%)                             872 non-null   float64
23  Population                                                           872 non-null   float64
24  Unemployment Rate                                                    872 non-null   float64
25  Urban Population                                                     872 non-null   float64
26  Latitude                                                             872 non-null   float64
27  Longitude                                                            872 non-null   float64
dtypes: float64(18), int64(3), object(7)
memory usage: 217.8+ KB
```

In [37]: *# Last 5 rows of dataset;*
df.tail()

Out[37]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Cou
990	991	Natan por Aĩç	12300000	9.029610e+09	Sports	Natan por Aĩç	1200	Bi
991	992	Free Fire India Official	12300000	1.674410e+09	People & Blogs	Free Fire India Official	1500	Il
992	993	Panda	12300000	2.214684e+09	NaN	HybridPanda	2452	Un Kingr
993	994	RobTopGames	12300000	3.741235e+08	Gaming	RobTopGames	39	Swe
994	995	Make Joke Of	12300000	2.129774e+09	Comedy	Make Joke Of	62	Il

5 rows × 28 columns

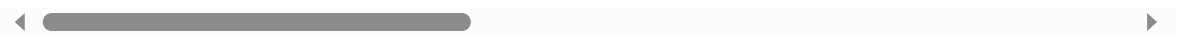


```
In [38]: # converting subscribers , video views and population, urban population in mi
df['Subscribers'] = (df['Subscribers'] / 1000000).round(2)
df['Video Views'] = (df['Video Views'] / 1000000).round(2)
df["Population"] = (df['Population'] / 1000000).round(2)
df['Urban Population'] = (df['Urban Population'] / 1000000).round(2)
df.tail()
```

Out[38]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Country
990	991	Natan por Aĩç	12.3	9029.61	Sports	Natan por Aĩç	1200	Brazil
991	992	Free Fire India Official	12.3	1674.41	People & Blogs	Free Fire India Official	1500	India
992	993	Panda	12.3	2214.68	NaN	HybridPanda	2452	United Kingdom
993	994	RobTopGames	12.3	374.12	Gaming	RobTopGames	39	Sweden
994	995	Make Joke Of	12.3	2129.77	Comedy	Make Joke Of	62	India

5 rows × 28 columns



```
In [39]: # checking of null values
df.isnull().sum()
```

```
Out[39]: Rank                                0
Youtuber                                    0
Subscribers                                0
Video Views                                0
Category                                   46
Title                                       0
Uploads                                    0
Country                                   122
Abbreviation                              122
Channel Type                               30
Video Views Rank                           1
Country Rank                               116
Channel Type Rank                           33
Video Views for the last 30 days             56
Lowest Monthly Earnings                     0
Highest Monthly Earnings                    0
Lowest Yearly Earnings                      0
Highest Yearly Earnings                     0
Subscribers for last 30 days                 337
Created Year                                 5
Created Month                               5
Created Date                                5
Gross Tertiary Education Enrollment (%)     123
Population                                 123
Unemployment Rate                           123
Urban Population                            123
Latitude                                    123
Longitude                                    123
dtype: int64
```

```
In [40]: df.shape
```

```
Out[40]: (995, 28)
```

Data Cleaning;

```
In [41]: # finding the duplicates values;  
df.duplicated().sum()
```

```
Out[41]: 0
```

```
In [42]: # finding number of unique rows;  
df[~df.duplicated()].shape[0]
```

```
Out[42]: 995
```

```
In [43]: # handling missing values in dataset  
NaNnumerical= df.select_dtypes(include=["float64","int64"] ).columns  
NaNcategorical= df.select_dtypes(include=["object"]).columns  
df[NaNnumerical]= df[NaNnumerical].fillna(0)  
df[NaNcategorical]= df[NaNcategorical].fillna("Unknown")
```

```
In [44]: df.isnull().sum()
```

```
Out[44]: Rank                                0  
Youtuber                                    0  
Subscribers                                0  
Video Views                                0  
Category                                    0  
Title                                       0  
Uploads                                    0  
Country                                    0  
Abbreviation                               0  
Channel Type                               0  
Video Views Rank                           0  
Country Rank                               0  
Channel Type Rank                           0  
Video Views for the last 30 days            0  
Lowest Monthly Earnings                     0  
Highest Monthly Earnings                    0  
Lowest Yearly Earnings                      0  
Highest Yearly Earnings                     0  
Subscribers for last 30 days                0  
Created Year                               0  
Created Month                              0  
Created Date                               0  
Gross Tertiary Education Enrollment (%)     0  
Population                                  0  
Unemployment Rate                          0  
Urban Population                           0  
Latitude                                    0  
Longitude                                   0  
dtype: int64
```

```
In [45]: # analysis of value in statistical format;
describe_df=df.describe()
print(describe_df)
```

	Rank	Subscribers	Video Views	Uploads	Video Views Ra
nk \					
count	995.00000	995.000000	995.000000	995.000000	9.950000e+
02					
mean	498.00000	22.982412	11039.536724	9187.125628	5.536919e+
05					
std	287.37606	17.526105	14110.844384	34151.352254	1.362210e+
06					
min	1.00000	12.300000	0.000000	0.000000	0.000000e+
00					
25%	249.50000	14.500000	4288.145000	194.500000	3.175000e+
02					
50%	498.00000	17.700000	7760.820000	729.000000	9.130000e+
02					
75%	746.50000	24.600000	13554.700000	2667.500000	3.579000e+
03					
max	995.00000	245.000000	228000.000000	301308.000000	4.057944e+
06					

	Country Rank	Channel Type Rank	Video Views for the last 30 days \
count	995.000000	995.000000	9.950000e+02
mean	341.046231	720.986935	1.657267e+08
std	1164.728018	1916.500613	4.065010e+08
min	0.000000	0.000000	0.000000e+00
25%	5.000000	24.000000	1.351700e+07
50%	34.000000	62.000000	5.635800e+07
75%	114.000000	137.000000	1.585655e+08
max	7741.000000	7741.000000	6.589000e+09

	Lowest Monthly Earnings	Highest Monthly Earnings	...	\
count	995.000000	9.950000e+02	...	
mean	36886.148281	5.898078e+05	...	
std	71858.724092	1.148622e+06	...	
min	0.000000	0.000000e+00	...	
25%	2700.000000	4.350000e+04	...	
50%	13300.000000	2.127000e+05	...	
75%	37900.000000	6.068000e+05	...	
max	850900.000000	1.360000e+07	...	

	Highest Yearly Earnings	Subscribers for last 30 days	Created Year
\			
count	9.950000e+02	9.950000e+02	995.000000
mean	7.081814e+06	2.308483e+05	2002.516583
std	1.379704e+07	5.261092e+05	142.455353
min	0.000000e+00	0.000000e+00	0.000000
25%	5.217500e+05	0.000000e+00	2009.000000
50%	2.600000e+06	1.000000e+05	2013.000000
75%	7.300000e+06	2.000000e+05	2016.000000
max	1.634000e+08	8.000000e+06	2022.000000

	Created Date	Gross Tertiary Education Enrollment (%)	Population
\			
count	995.000000	995.000000	995.000000
mean	15.667337	55.762211	377.183980
std	8.826000	32.191178	464.717802
min	0.000000	0.000000	0.000000
25%	8.000000	28.100000	50.340000
50%	16.000000	60.000000	270.200000
75%	23.000000	88.200000	328.240000
max	31.000000	113.100000	1397.720000

	Unemployment Rate	Urban Population	Latitude	Longitude
count	995.000000	995.000000	995.000000	995.000000
mean	8.132191	196.496844	23.340489	-12.381652
std	5.502432	162.538385	21.150562	79.479723
min	0.000000	0.000000	-38.416097	-172.104629
25%	3.850000	40.830000	4.570868	-95.712891
50%	5.360000	183.240000	23.634501	-3.435973
75%	14.700000	270.660000	37.090240	78.962880
max	14.720000	842.930000	61.924110	138.252924

[8 rows x 21 columns]

Data manipulation;

```
In [46]: # rename the column name ;
df.rename(columns={"Unemployment rate": "Unemployment rate(%)"}, inplace=True)

# Display the modified DataFrame
df.head()
```

Out[46]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Country
0	1	T-Series	245.0	228000.00	Music	T-Series	20082	India
1	2	YouTube Movies	170.0	0.00	Film & Animation	youtubemovies	1	United States
2	3	MrBeast	166.0	28368.84	Entertainment	MrBeast	741	United States
3	4	Cocomelon - Nursery Rhymes	162.0	164000.00	Education	Cocomelon - Nursery Rhymes	966	United States
4	5	SET India	159.0	148000.00	Shows	SET India	116536	India

5 rows x 28 columns



```
In [47]: # filter the rows where the videos views are not equal to 0;
df[df['Video Views']!=0]
```

Out[47]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Cou
0	1	T-Series	245.0	228000.00	Music	T-Series	20082	I
2	3	MrBeast	166.0	28368.84	Entertainment	MrBeast	741	Ur St
3	4	Cocomelon - Nursery Rhymes	162.0	164000.00	Education	Cocomelon - Nursery Rhymes	966	Ur St
4	5	SET India	159.0	148000.00	Shows	SET India	116536	I
6	7	ýýý Kids Diana Show	112.0	93247.04	People & Blogs	ýýý Kids Diana Show	1111	Ur St
...	...	...	...	...	...	...	...	
990	991	Natan por Aĩž	12.3	9029.61	Sports	Natan por Aĩž	1200	B
991	992	Free Fire India Official	12.3	1674.41	People & Blogs	Free Fire India Official	1500	I
992	993	Panda	12.3	2214.68	Unknown	HybridPanda	2452	Ur King
993	994	RobTopGames	12.3	374.12	Gaming	RobTopGames	39	Swe
994	995	Make Joke Of	12.3	2129.77	Comedy	Make Joke Of	62	I

986 rows × 28 columns



```
In [48]: # creating new column in dataset named 'AvarageYearlyEarnings'
df['AverageYearlyEarnings'] = (df['Lowest Yearly Earnings'] + df['Highest Ye
df['AverageYearlyEarnings'] = (df['AverageYearlyEarnings'] / 1000000).round(
df.head())
```

Out[48]:

	Rank	Youtuber	Subscribers	Video Views	Category	Title	Uploads	Country
0	1	T-Series	245.0	228000.00	Music	T-Series	20082	India
1	2	YouTube Movies	170.0	0.00	Film & Animation	youtubemovies	1	United States
2	3	MrBeast	166.0	28368.84	Entertainment	MrBeast	741	United States
3	4	Cocomelon - Nursery Rhymes	162.0	164000.00	Education	Cocomelon - Nursery Rhymes	966	United States
4	5	SET India	159.0	148000.00	Shows	SET India	116536	India

5 rows × 29 columns



Data analysis and visualization;

Company wants to run advertisement in the country with maximum subscribers.

Question 1: Find out which country has most subscribers?

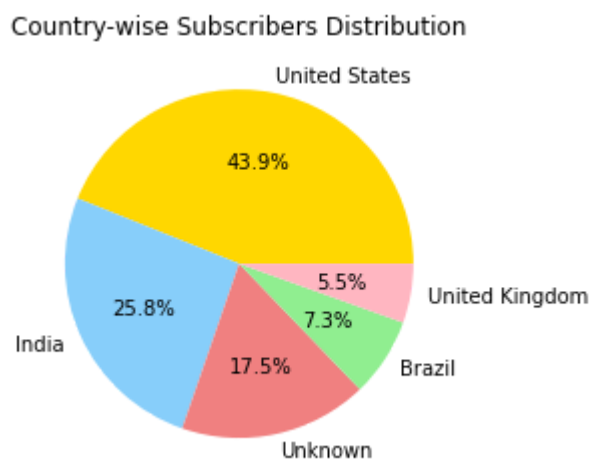
Insight: The distribution of country-wise subscribers reveals that the United States has the highest percentage of subscribers at 43.9%, followed by India at 25.8%, Unknown regions at 17.5%, Brazil at 7.3%, and the United Kingdom at 5.5%.

Root Cause: Variations in subscriber percentages can be attributed to factors such as population size, internet penetration, cultural interests, and the popularity of YouTube content in each country.

Provable Solution: Localized Content Develop Create content that aligns with the cultural preferences and interests of the dominant market.

```
In [49]: country_subscribers = df.groupby('Country')['Subscribers'].sum().sort_values
plt.pie(country_subscribers, labels=country_subscribers.index, autopct='%1.1f%%',
        colors=['gold', 'lightskyblue', 'lightcoral', 'lightgreen', 'lightpink'])

plt.title('Country-wise Subscribers Distribution')
plt.show()
```



Question 2: Find out top 10 Indian Channels having most subscribers?

Insights: T-Series leads with the highest YouTube subscribers in India, followed by entertainment and music channels like SET India and Zee Music Company.

Root Cause: Dominance of music and entertainment content suggests a strong preference for these genres among Indian viewers.

Provable Solution: Content creators targeting the Indian audience should consider producing music and entertainment content to tap into the popularity of these genres and potentially

```
In [50]: # find the top 10 indian youtube subscribers;
indian_channels = df[df['Abbreviation'] == 'IN']
top_10_indian_channels = indian_channels.sort_values(by='Subscribers', ascending=False)

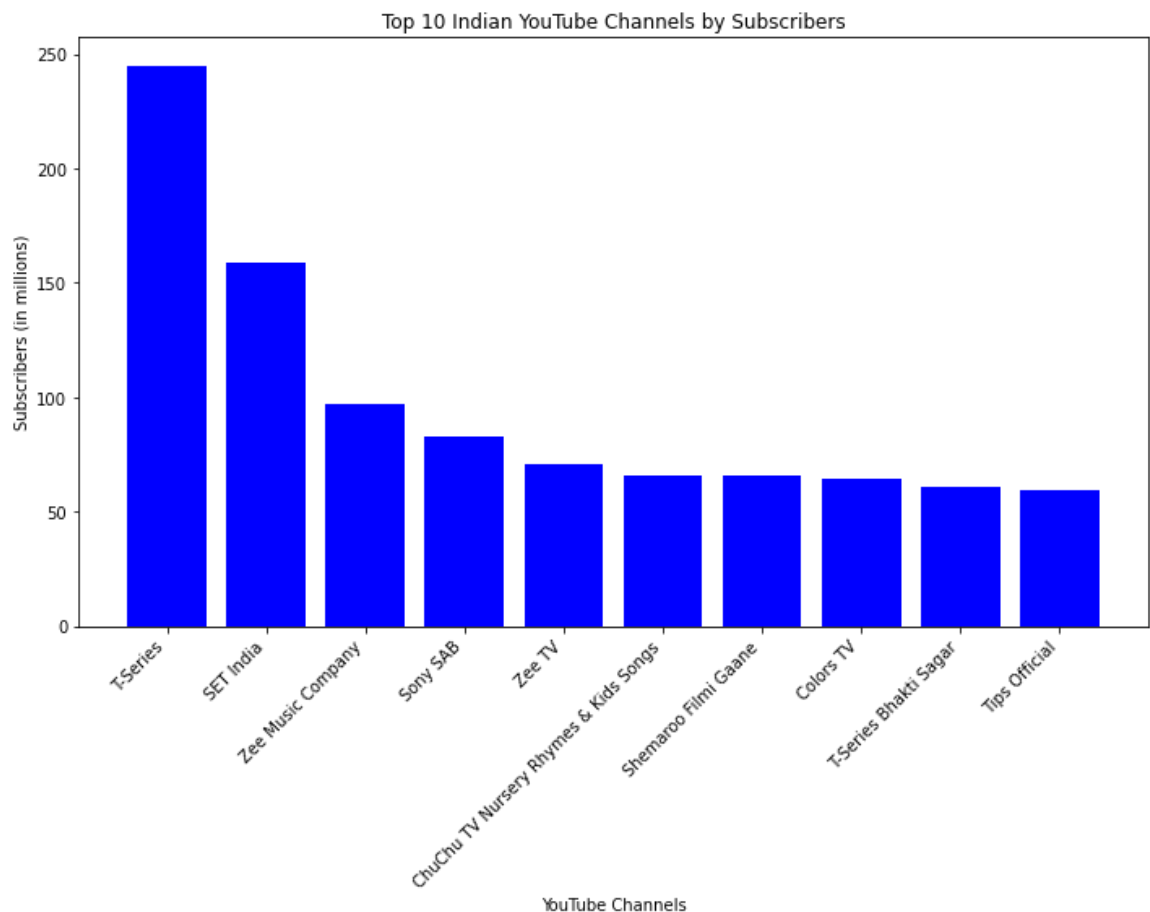
print(top_10_indian_channels[['Youtuber', 'Subscribers']])

import matplotlib.pyplot as plt

plt.figure(figsize=(10, 8))
plt.bar(top_10_indian_channels['Youtuber'], top_10_indian_channels['Subscribers'])
plt.xlabel('YouTube Channels')
plt.ylabel('Subscribers (in millions)')
plt.title('Top 10 Indian YouTube Channels by Subscribers')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()
```

	Youtuber	Subscribers
0	T-Series	245.0
4	SET India	159.0
10	Zee Music Company	96.7
15	Sony SAB	83.0
21	Zee TV	70.5
24	ChuChu TV Nursery Rhymes & Kids Songs	65.9
25	Shemaroo Filmi Gaane	65.6
26	Colors TV	64.6
27	T-Series Bhakti Sagar	61.0
30	Tips Official	59.3



Question 3: Find out percentage of each Category out of Total subscribers?

Insight 3: Music and Entertainment categories have the highest percentage of subscribers, indicating a strong audience preference for these genres.

Root Cause: The popularity of music and entertainment content may be due to their universal appeal and ability to engage a wide audience.

Provable Solution: Content creators should strategically incorporate music and entertainment elements into their videos aligning with the preferences of the majority of subscribers, to maximize engagement and subscriber growth.

```
In [51]: # category wise subscribers in percentage in horizontal bar chart;
import matplotlib.pyplot as plt
import seaborn as sns

category_subscribers = df.groupby('Category')['Subscribers'].sum()
total_subscribers = df['Subscribers'].sum()

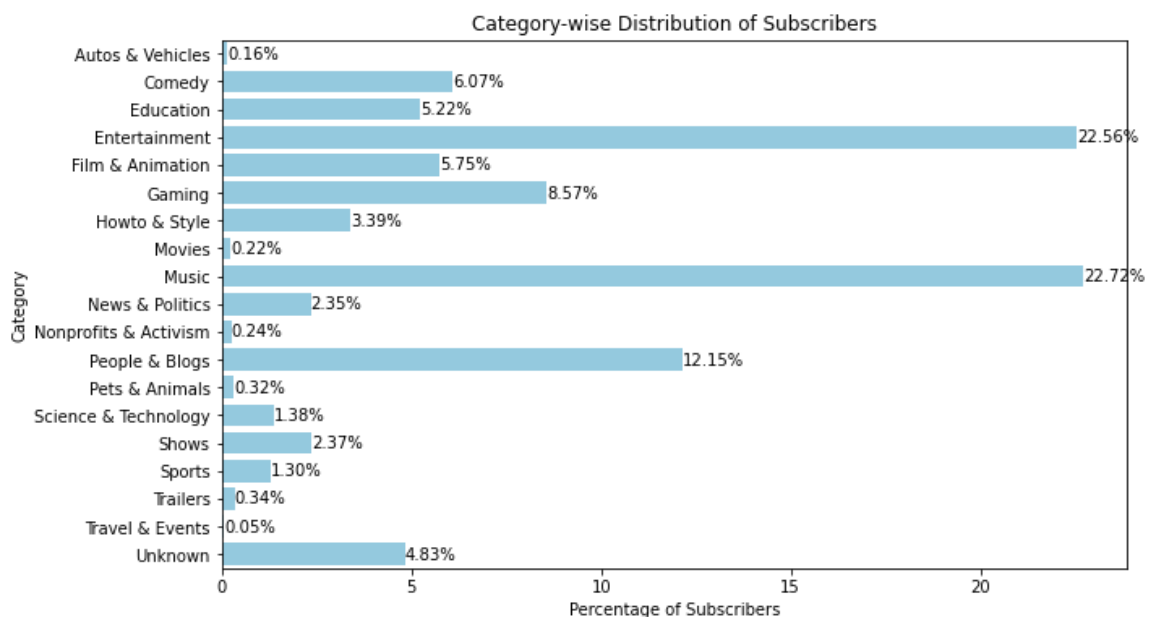
percentage_subscribers = (category_subscribers / total_subscribers) * 100
percentage_subscribers = percentage_subscribers.reset_index()

plt.figure(figsize=(10, 6))
bar_plot = sns.barplot(x='Subscribers', y='Category', data=percentage_subscribers)

for index, value in enumerate(percentagesubscribers['Subscribers']):
    bar_plot.text(value, index, f'{value:.2f}%', va='center')

plt.xlabel('Percentage of Subscribers')
plt.ylabel('Category')
plt.title('Category-wise Distribution of Subscribers')

plt.show()
```



Question 4: Comparison between India and US with Entertainment and Music category popularity?

Dominance of Music in India: The top 5 YouTubers in the Music category have a significant share of subscribers in India and US, Music content resonates strongly with the Indian audience.

Root Cause: Cultural Preference for Music India has a rich cultural heritage with a strong emphasis on music and entertainment.

Factors: Cultural influence, diverse music genres, and widespread music appreciation.

Provable Solution: Strategy: Content creators targeting the Indian audience should align with the cultural preference for music.

Action: Explore collaborations with musicians, cover popular songs, and produce content that integrates cultural elements to maximize engagement and subscriber growth. Both Entertainment and Music categories have diverse genres represented in the top 5 YouTubers in the US. US audiences have varied interests across different content genres.

Root cause: Hollywood influence, celebrity culture, and the prominence of entertainment content.

Provable Solution: Collaborate with influencers, leverage popular trends, and create engaging and shareable content to maintain and grow the subscriber base. Content creators targeting the US audience should embrace diversity and personalize content.

```

In [52]: filtered_df = df[df['Category'].isin(['Entertainment', 'Music']) & df['Abbre

india_entertainment = filtered_df[(filtered_df['Abbreviation'] == 'IN') & (f
india_music = filtered_df[(filtered_df['Abbreviation'] == 'IN') & (filtered_
us_entertainment = filtered_df[(filtered_df['Abbreviation'] == 'US') & (filt
us_music = filtered_df[(filtered_df['Abbreviation'] == 'US') & (filtered_df[

def get_top_5(dataframe):
    return dataframe.nlargest(5, 'Subscribers')[['Youtuber', 'Subscribers',

top_5_india_entertainment = get_top_5(india_entertainment)
top_5_india_music = get_top_5(india_music)
top_5_us_entertainment = get_top_5(us_entertainment)
top_5_us_music = get_top_5(us_music)

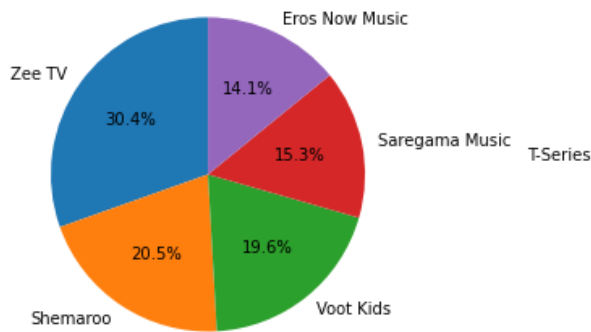
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(10, 10))
def plot_pie(ax, labels, sizes, title):
    ax.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90)
    ax.axis('equal')
    ax.set_title(title)

plot_pie(axes[0, 0], top_5_india_entertainment['Youtuber'], top_5_india_ente
plot_pie(axes[0, 1], top_5_india_music['Youtuber'], top_5_india_music['Subsc
plot_pie(axes[1, 0], top_5_us_entertainment['Youtuber'], top_5_us_entertainm
plot_pie(axes[1, 1], top_5_us_music['Youtuber'], top_5_us_music['Subscribers

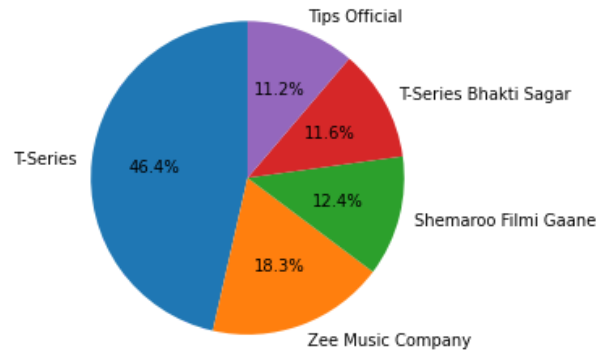
plt.tight_layout()
plt.show()

```

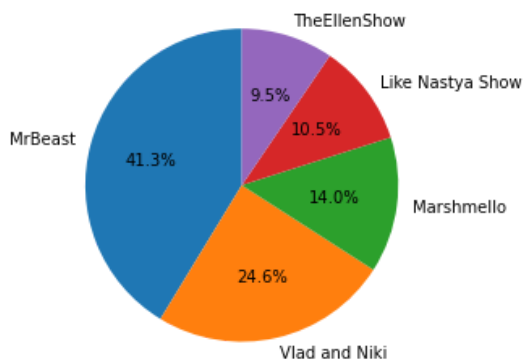
Top 5 Entertainment YouTubers (India)



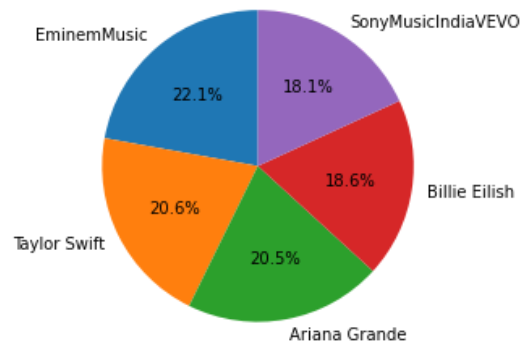
Top 5 Music YouTubers (India)



Top 5 Entertainment YouTubers (US)



Top 5 Music YouTubers (US)



Question 5: Give list of Top 10 YouTubers by highest average yearly earnings for giving gift hampers?

Insight: KIMPRO tops the list with the highest average yearly earnings, followed by Boom and T-Series, indicating varied revenue streams and successful monetization strategies.

Root Cause: Diverse revenue sources, including sponsorships, merchandise, and affiliate marketing, contribute to the high average earnings.

Provable Solution: Content creators should explore multiple revenue streams beyond ad revenue, such as collaborations, merchandise sales, and affiliate partnerships, to optimize their overall yearly earnings.

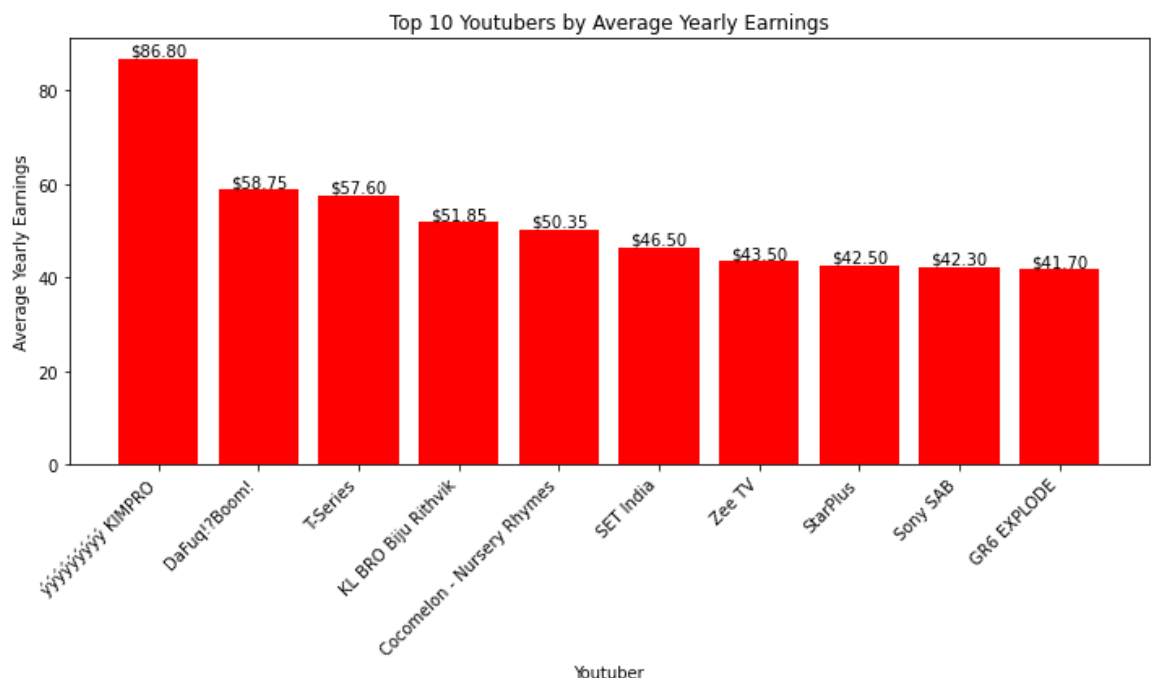
```
In [53]: # Top 10 youtubers by average yearly earnings:
import matplotlib.pyplot as plt

temp = df[['Youtuber', 'AverageYearlyEarnings']].sort_values(by='AverageYearlyEarnings', ascending=False)

plt.figure(figsize=(10, 6))
plt.bar(temp['Youtuber'], temp['AverageYearlyEarnings'], color='red')
plt.xlabel('Youtuber')
plt.ylabel('Average Yearly Earnings')
plt.title('Top 10 Youtubers by Average Yearly Earnings')
plt.xticks(rotation=45, ha='right')

# Labels
for index, value in enumerate(temp['AverageYearlyEarnings']):
    plt.text(index, value, f"${value:.2f}", ha='center', va='bottom')

plt.tight_layout()
plt.show()
```



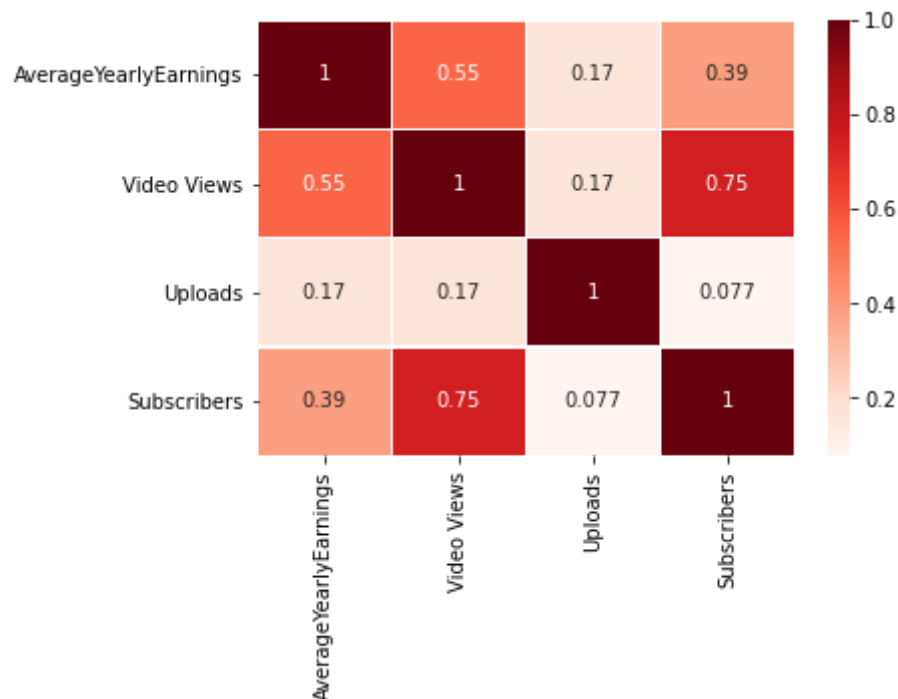
Question 6: Correlation of Video Views, Yearly earnings and Subscriber.

Insight: KIMPRO tops the list with the highest average yearly earnings, followed by Boom and T-Series, indicating varied revenue streams and successful monetization strategies.

Root Cause: Diverse revenue sources, including sponsorships, merchandise, and affiliate marketing, contribute to the high average earnings.

Provable Solution: Content creators should explore multiple revenue streams beyond ad revenue, such as collaborations, merchandise sales, and affiliate partnerships, to optimize their overall yearly earnings.

```
In [54]: df_corr = df[['AverageYearlyEarnings', 'Video Views', 'Uploads', 'Subscribers']
corr = df_corr.corr()
sns.heatmap(corr, annot=True, linewidth=0.5, cmap='Reds')
plt.show()
```



Question 7: Find out which category of channels are most viewed. Give Top 5 Channel type with their percentage?

Insight: The pie chart visually represents the distribution of Top 5 channel types in the countries by population, revealing the dominant channel types in these high-population regions.

Root Cause: Variances in channel type distribution may be influenced by cultural preferences, regional trends, or content consumption habits in countries with higher populations.

Provable Solution: Tailor content strategies based on the dominant channel types in each country, adapting to local preferences and trends to maximize audience engagement and subscriber growth.

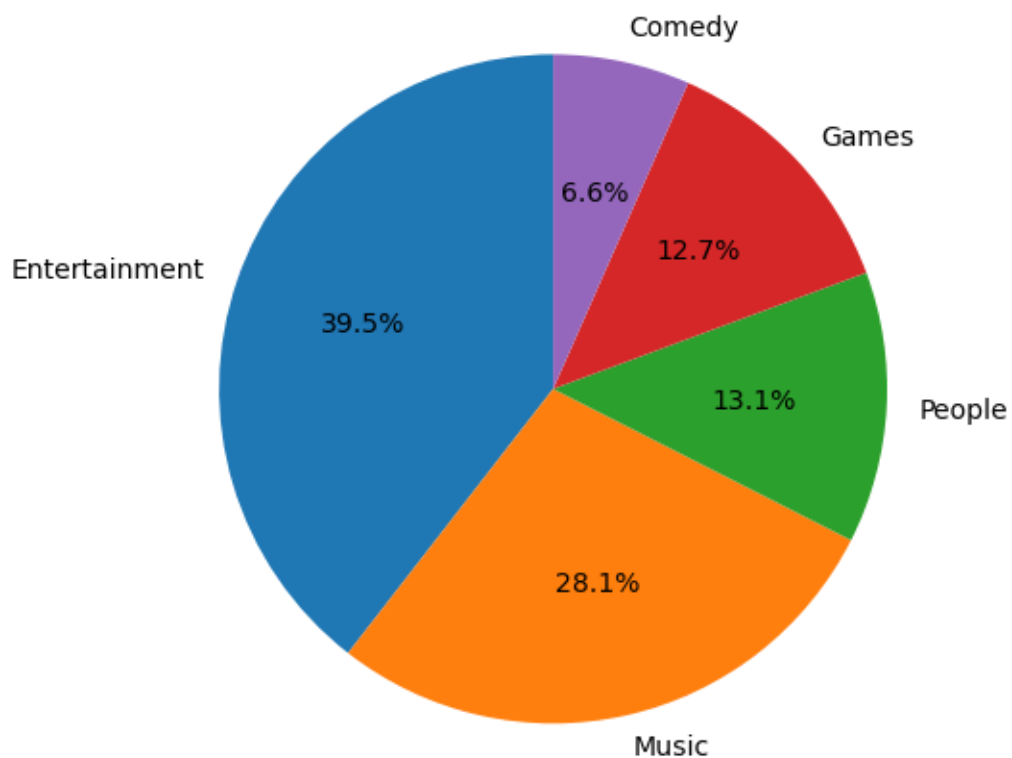
```
In [55]: top_countries = df.groupby('Country')['Population'].max().sort_values(ascending=False)
print(f'Countries with Highest Population: {top_countries}')

plt.figure(figsize=(20, 8))
channel_type_counts_top5 = df[df['Country'].isin(top_countries)]['Channel Type'].value_counts()

plt.pie(channel_type_counts_top5, labels=channel_type_counts_top5.index, autopct='%1.1f%%',
plt.title('Top 5 Channel Type Distribution in Countries by Population', font
plt.show()
```

Countries with Highest Population: ['China', 'India', 'United States', 'Indonesia', 'Pakistan', 'Brazil', 'Bangladesh', 'Russia', 'Japan', 'Mexico', 'Philippines', 'Egypt', 'Vietnam', 'Turkey', 'Germany', 'Thailand', 'France', 'United Kingdom', 'Italy', 'South Korea', 'Colombia', 'Spain', 'Argentina', 'Ukraine', 'Iraq', 'Afghanistan', 'Canada', 'Morocco', 'Saudi Arabia', 'Peru', 'Malaysia', 'Venezuela', 'Australia', 'Chile', 'Ecuador', 'Netherlands', 'Cuba', 'Sweden', 'Jordan', 'United Arab Emirates', 'Switzerland', 'El Salvador', 'Singapore', 'Finland', 'Kuwait', 'Latvia', 'Barbados', 'Samoa', 'Andorra', 'Unknown']

Top 5 Channel Type Distribution in Countries by Population



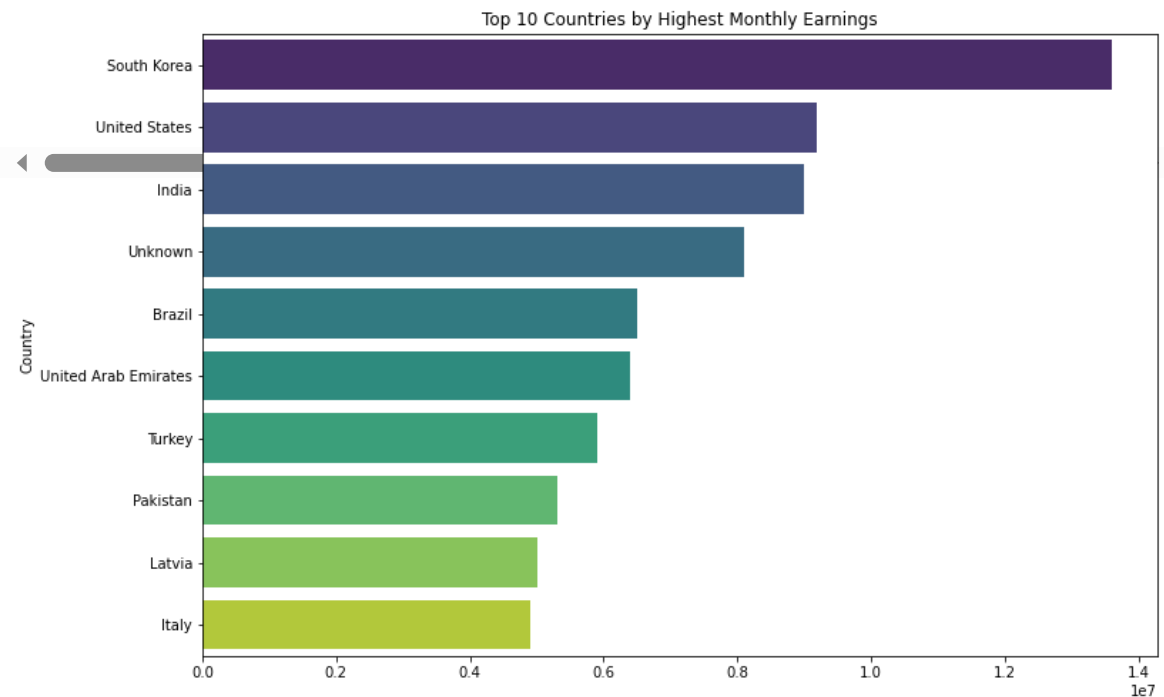
Question 8: Which country earns the most from Youtube. Give list of top 10 countries?

Insight: The top earners are not concentrated in a specific region, indicating a global distribution of successful YouTube content creators.

Root Cause: Variances in the availability of monetization opportunities, such as ad revenue, sponsorships, and partnerships.

Provable Solution: Encourage content creators to tailor their content to local preferences and cultures. Provide resources and guidance to creators on understanding and adapting content to specific regional interests.

```
In [56]: top_countries = df.groupby('Country')['Highest Monthly Earnings'].max().sort
plt.figure(figsize=(12, 8))
sns.barplot(x=top_countries.values, y=top_countries.index, palette='viridis')
plt.title('Top 10 Countries by Highest Monthly Earnings')
plt.show()
```



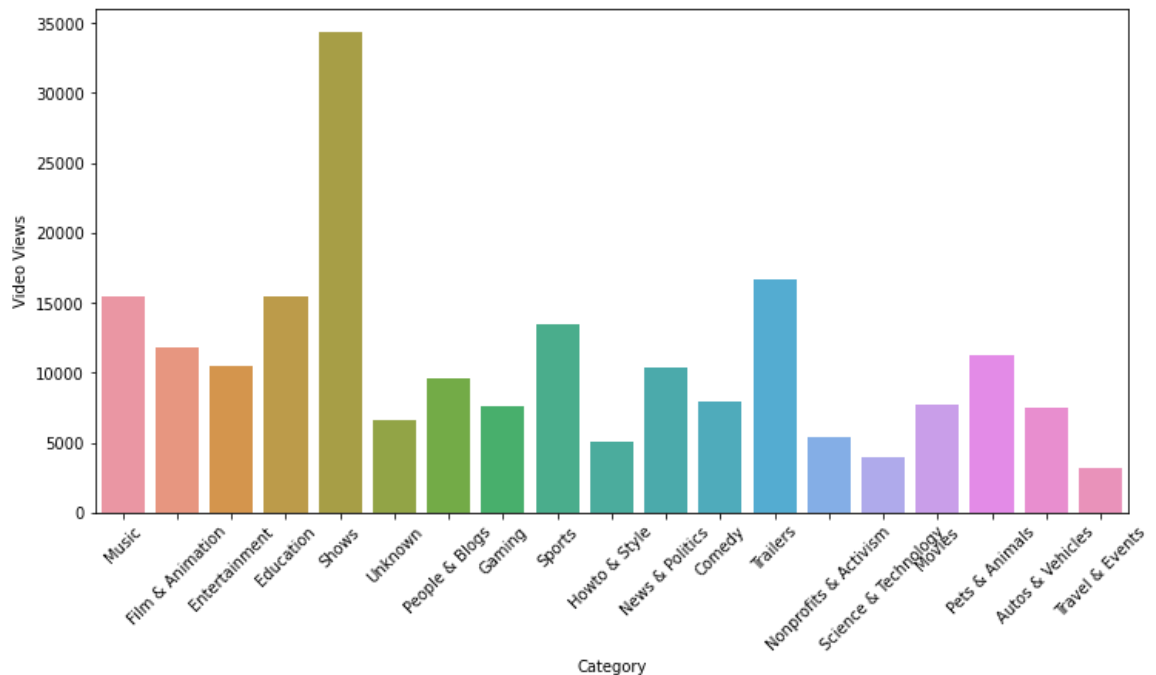
Question 9: Video Views Distribution by Channel Category?

Insight: The bar plot illustrates the distribution of video views across different content categories, highlighting variations in popularity.

Root Cause: Differences in video views by category may stem from varying audience interests, content quality, or trends in specific categories.

Provable Solution: Content creators should analyze the performance of each category, identify popular trends, and strategically align content creation with high-performing categories to enhance video views and audience engagement. Consistent monitoring of category-wise performance can guide content strategy adjustments.

```
In [76]: plt.figure(figsize=(12, 6))
sns.barplot(x='Category', y='Video Views', data=df, ci=None)
plt.xticks(rotation=45)
plt.show()
```



Question 10: Make code of relation regression line for subscribers and video views?

Insights: As the number of subscribers increases, there is a tendency for video views to also increase.

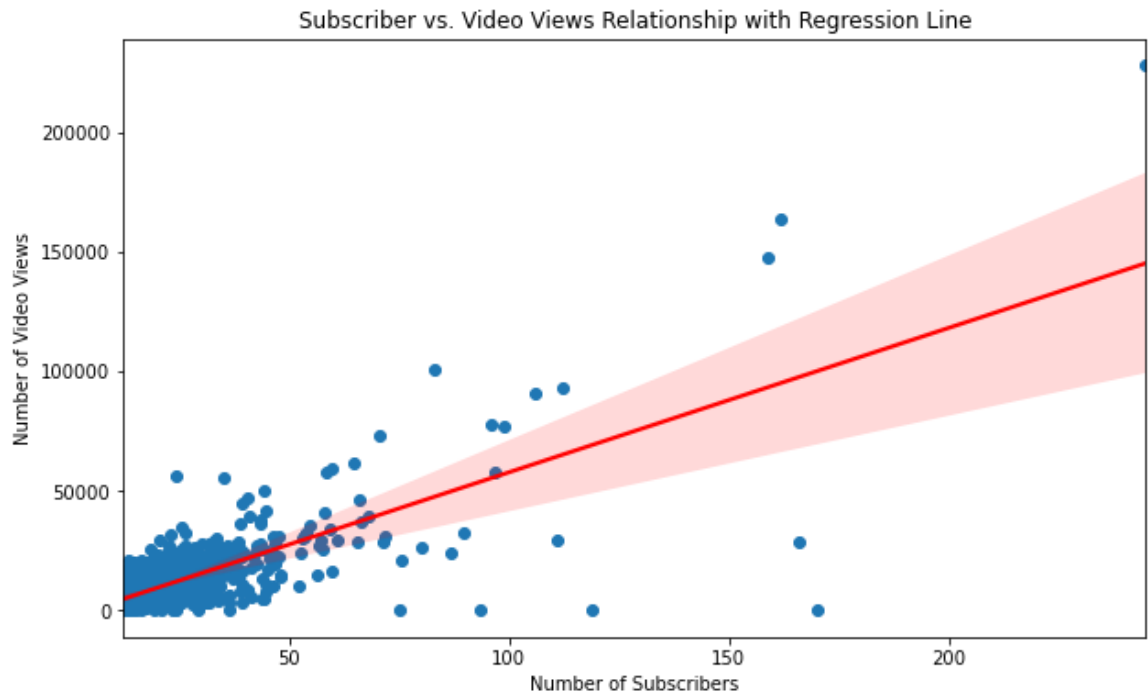
Root Cause: Engaged subscribers are more likely to watch a creator's videos regularly, contributing to increased video views. Content creators should prioritize building and maintaining a loyal subscriber base.

Provable Solution: Encouraging user interactions, such as likes, comments, and shares, can further enhance subscriber engagement.

```
In [75]: plt.figure(figsize=(10, 6))
sns.regplot(x='Subscribers', y='Video Views', data=df, scatter_kws={'alpha':

plt.title('Subscriber vs. Video Views Relationship with Regression Line')
plt.xlabel('Number of Subscribers')
plt.ylabel('Number of Video Views')

plt.show()
```



Question 11: Find the category with the maximum occurrence in India ?

Insights: The bar plot reveals that the majority of YouTube channels in India fall under the "Entertainment" category, indicating a significant interest in diverse entertainment content among the Indian audience. While Entertainment is dominant, the presence of other categories. The presence of the "Education" category in the top 5 indicates a growing interest in educational content, presenting an opportunity for content creators to focus on providing educational videos.

Root Cause: The dominance of the "Entertainment" category may be rooted in cultural preferences, indicating a strong interest in music, movies, and general entertainment content. The inclusion of the "Education" category may be influenced by a growing trend in online learning and the increasing demand for educational content.

Provable Solution: Content creators can explore opportunities to diversify their content within popular categories. Channels in the "Education" category can explore partnerships with educational institutions or subject matter experts to enhance the quality and credibility of their content.

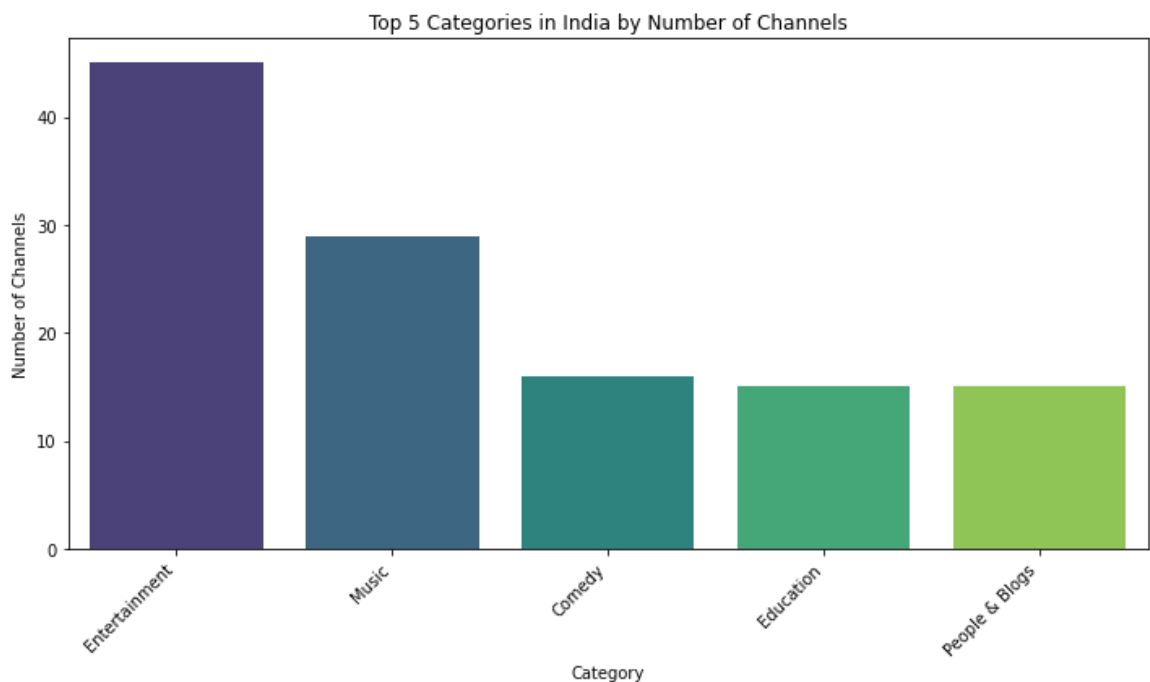
```
In [60]: india_data = df[df['Country'] == 'India']

top5_categories = india_data['Category'].value_counts().head(5)

plt.figure(figsize=(10, 6))
sns.barplot(x=top5_categories.index, y=top5_categories.values, palette='viri

plt.title('Top 5 Categories in India by Number of Channels')
plt.xlabel('Category')
plt.ylabel('Number of Channels')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()
```



Question 12: Find Comparison of Subscriber and Video Views?

Insight: There is a positive correlation between the number of subscribers and video views, suggesting that channels with more subscribers tend to attract higher viewership.

Root Cause: Engaged subscribers are more likely to watch videos regularly, contributing to increased video views.

Provable Solution: Focus on building a loyal subscriber base through consistent and engaging content, as it directly influences higher video views. Encourage user interactions, such as likes, comments, and shares, to enhance subscriber engagement.

```
In [61]: plt.figure(figsize=(12, 6))

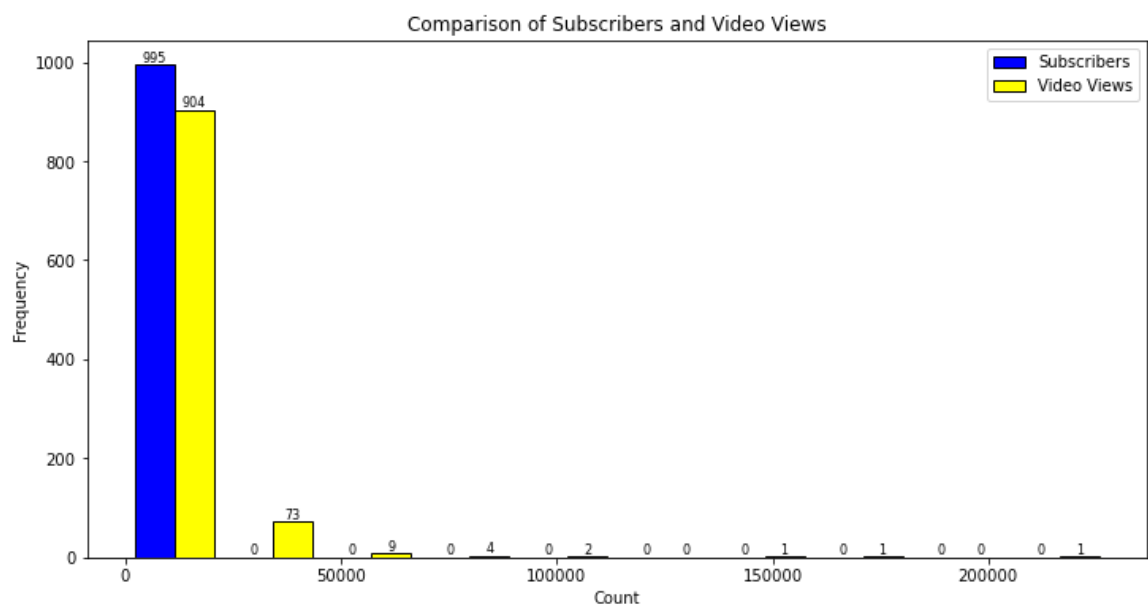
_, _, patches = plt.hist([df['Subscribers'], df['Video Views']], bins=10, co

for i in range(len(patches[0])):
    plt.text(patches[0][i].get_x() + patches[0][i].get_width() / 2, patches[
        str(int(patches[0][i].get_height()))), ha='center', va='bottom',

for i in range(len(patches[1])):
    plt.text(patches[1][i].get_x() + patches[1][i].get_width() / 2, patches[
        str(int(patches[1][i].get_height()))), ha='center', va='bottom',

# Histogram
plt.xlabel('Count')
plt.ylabel('Frequency')
plt.title('Comparison of Subscribers and Video Views')
plt.legend()

plt.show()
```



Question 13: No. of youtube channels from each category and country ?

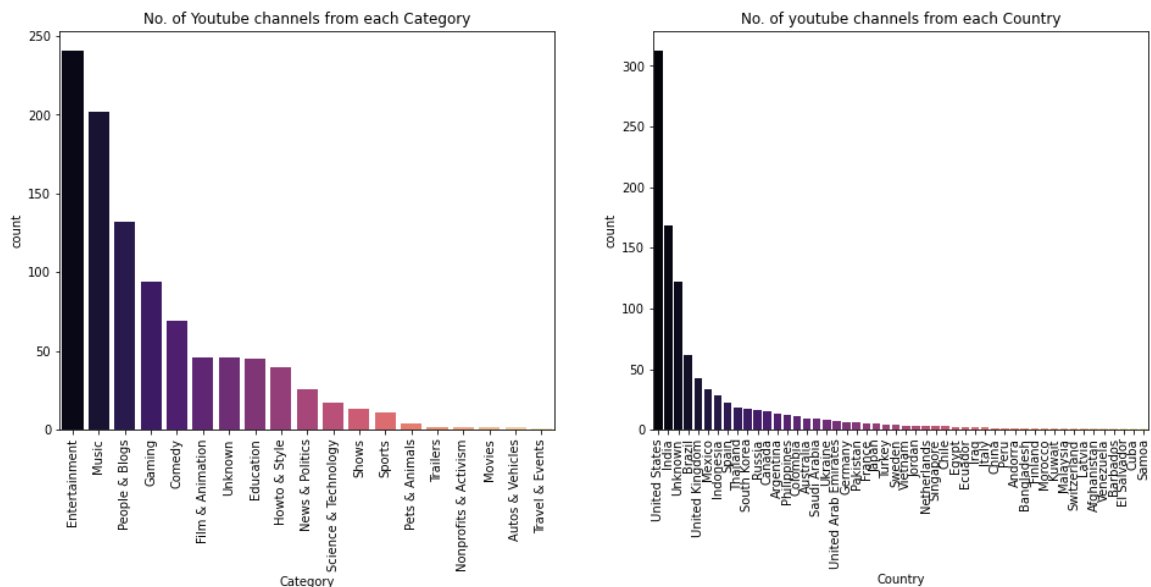
Insight: The count plots reveal that certain categories and countries dominate YouTube channels, indicating popular content and geographical concentrations.

Root Cause: Variances in the number of channels across categories and countries may result from cultural preferences, regional trends, and audience demographics.

Provable Solution: Content creators should analyze popular categories and tailor content to meet specific country preferences, potentially expanding into underrepresented categories and countries to diversify their audience and reach a broader market.

```
In [62]: fig,ax=plt.subplots(nrows=1,ncols=2,figsize=(16,6))
sns.countplot(x='Category',data=df,order=df['Category'].value_counts(ascending=True))
sns.countplot(x='Country',data=df,order=df['Country'].value_counts(ascending=True))
ax[0].set_xticklabels(list(df['Category'].value_counts(ascending=False).index))
ax[1].set_xticklabels(list(df['Country'].value_counts(ascending=False).index))
ax[0].set_title('No. of Youtube channels from each Category')
ax[1].set_title('No. of youtube channels from each Country')
```

Out[62]: Text(0.5, 1.0, 'No. of youtube channels from each Country')

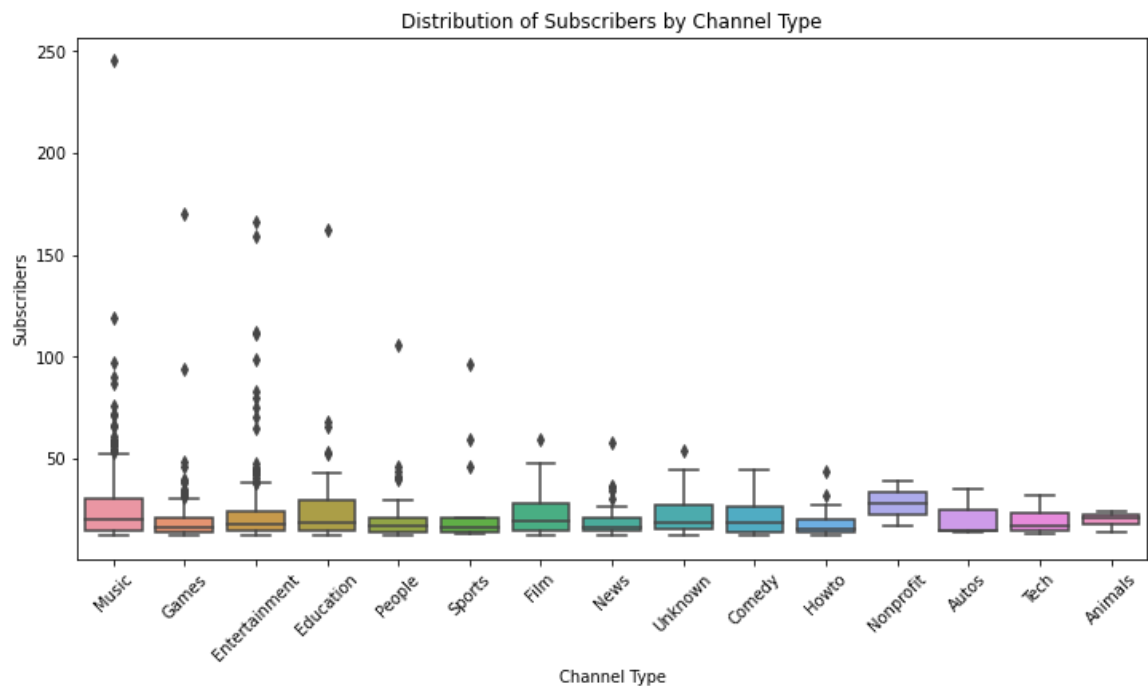


Insights 14: Different channel types exhibit varied distributions of subscribers.

Root cause: The type of content offered by different channel types influences the size and nature of their subscriber base. content format, and audience preferences.

Provable solution: Analyze successful channels within each type, identify content trends, and strategically position content to maximize subscriber growth.

```
In [63]: plt.figure(figsize=(12, 6))
sns.boxplot(x='Channel Type', y='Subscribers', data=df)
plt.title('Distribution of Subscribers by Channel Type')
plt.xticks(rotation=45)
plt.show()
```



Question 15: How many channels are created in years?

Insights : Understanding trends in channel creation over the years is crucial for assessing the platform's growth and content creator engagement. Peaks may indicate periods of increased interest, creator onboarding.

Root cause: Platform updates, algorithm changes, and shifts in user behavior. Peaks may be linked to influxes of new creators during specific periods.

Provable Solution: Provide continuous educational resources, community support, and opportunities for collaboration to maintain a healthy and stable creator ecosystem.

```
In [ ]: df['Created Year'] = df['Created Year'].astype(int)
df_filtered = df[df['Created Year'] != 1970]

year_counts = df_filtered['Created Year'].value_counts().sort_index()

plt.figure(figsize=(12, 6))
sns.barplot(x=year_counts.index, y=year_counts.values, color='blue')

plt.title('YouTube Channels Created Yearly Comparison')
plt.xlabel('Year')
plt.ylabel('Number of Channels')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()
```

16

```
In [74]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Assuming df is your DataFrame with 'Country', 'Unemployment Rate', and 'Ch

# Filter relevant columns
employment_channels_data = df[['Country', 'Unemployment Rate']].dropna()
channel_counts_by_country = df['Country'].value_counts().reset_index()
channel_counts_by_country.columns = ['Country', 'Channel Count']

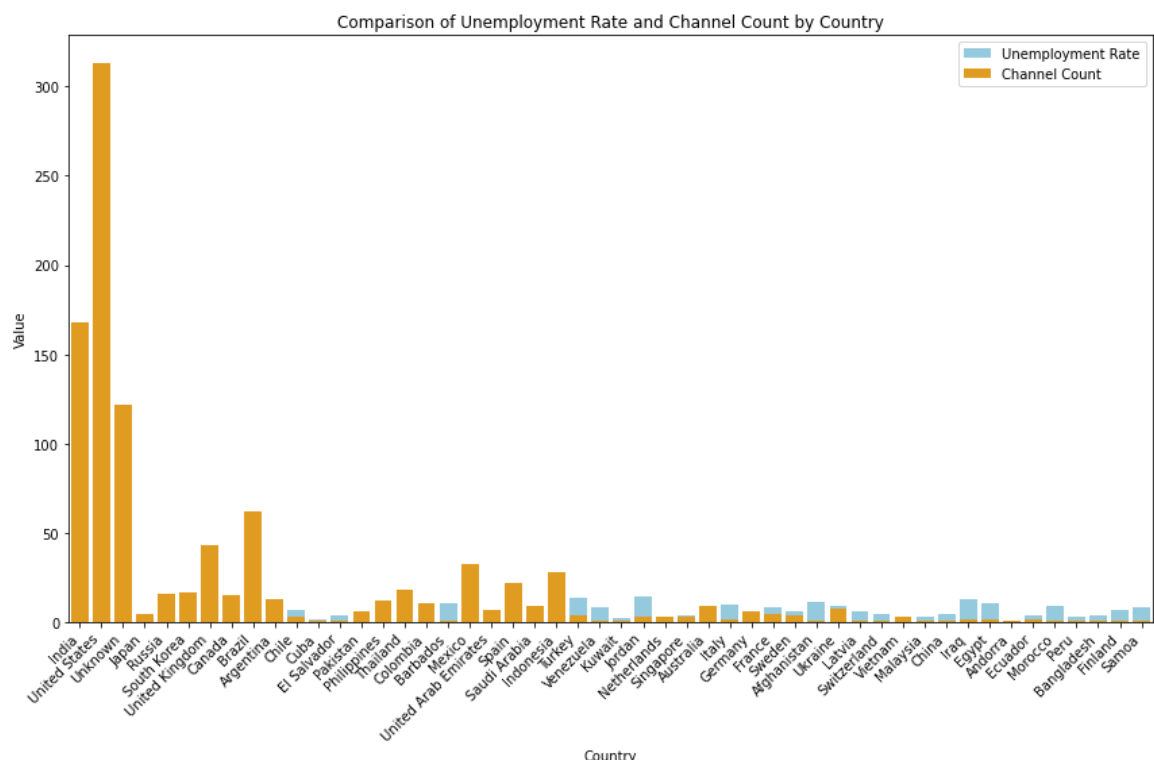
merged_data = pd.merge(employment_channels_data, channel_counts_by_country,

# Plotting
plt.figure(figsize=(12, 8))
sns.barplot(x='Country', y='Unemployment Rate', data=merged_data, color='sky
sns.barplot(x='Country', y='Channel Count', data=merged_data, color='orange'

plt.title('Comparison of Unemployment Rate and Channel Count by Country')
plt.xlabel('Country')
plt.ylabel('Value')
plt.legend()

plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()
```



17:

```

In [68]: employment_channels_data = df[['Country', 'Unemployment Rate']].dropna()
channel_counts_by_country = df['Country'].value_counts().reset_index()
channel_counts_by_country.columns = ['Country', 'Channel Count']

top10_countries = channel_counts_by_country.tail(10)['Country']

merged_data_top10 = merged_data[merged_data['Country'].isin(top10_countries)]

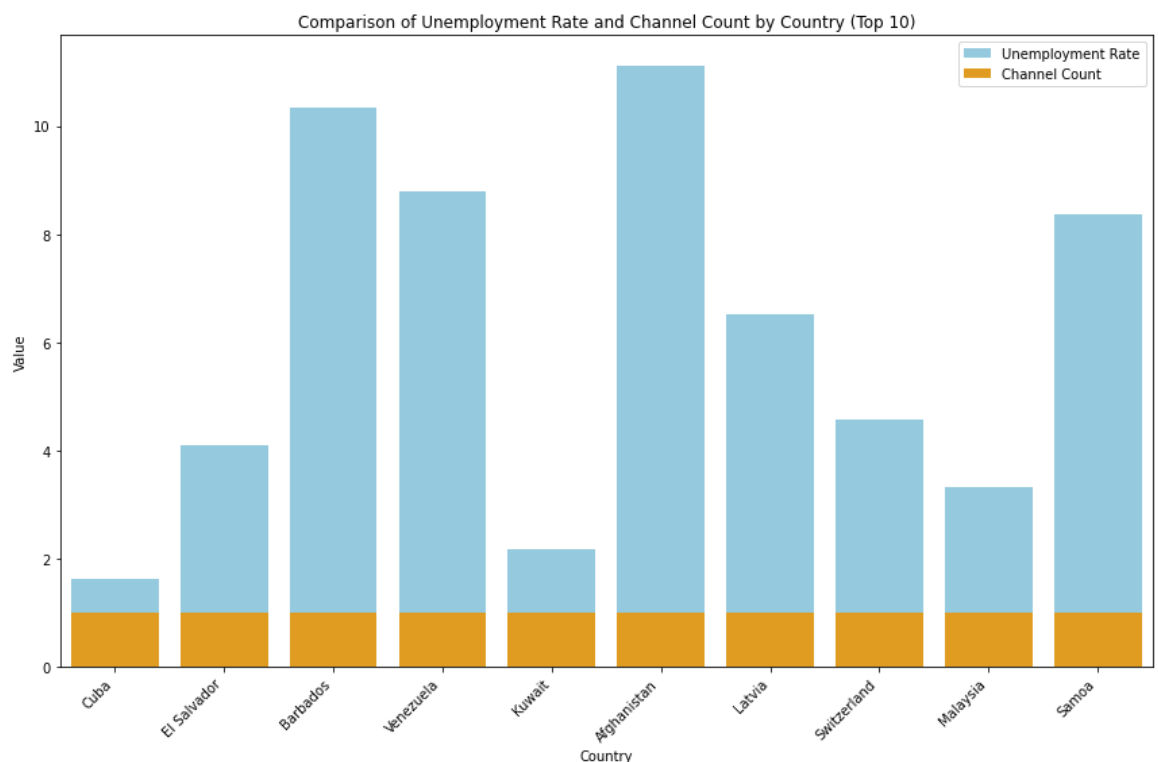
# Plotting
plt.figure(figsize=(12, 8))
sns.barplot(x='Country', y='Unemployment Rate', data=merged_data_top10, color='lightblue')
sns.barplot(x='Country', y='Channel Count', data=merged_data_top10, color='orange')

plt.title('Comparison of Unemployment Rate and Channel Count by Country (Top 10)')
plt.xlabel('Country')
plt.ylabel('Value')
plt.legend()

plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()

```

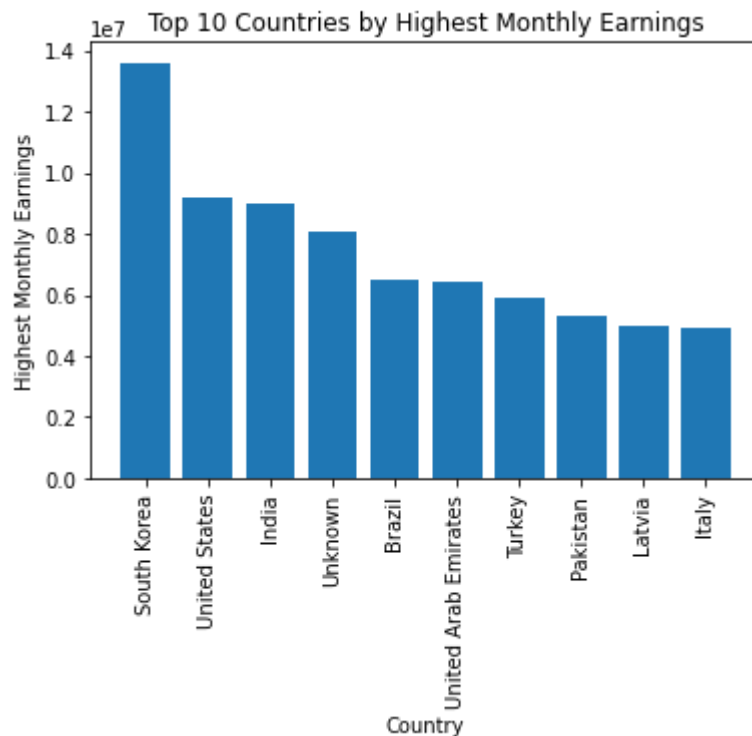


18:

```
In [65]: # Group the data by country and sort by highest monthly earnings
grouped = df.groupby('Country')['Highest Monthly Earnings'].max().sort_values

top10 = grouped.head(10)

# Create a bar chart of the top 10 countries
plt.bar(top10.index, top10.values)
plt.xticks(rotation=90)
plt.xlabel('Country')
plt.ylabel('Highest Monthly Earnings')
plt.title('Top 10 Countries by Highest Monthly Earnings')
plt.show()
```



19:

```
In [66]: # Filter rows where Country is India
india_data = df[df['Country'] == 'India']

max_category = india_data['Category'].value_counts().idxmax()

print(f"The maximum category for India is: {max_category}")
```

The maximum category for India is: Entertainment